Homework 6: Bootstrap methods, MCMC, and variational inference

Submission

Due: Nov 23rd, 2026

Submit the following three files in your directory:

1. `lastname_firstname_hw06.ipynb`: one executed Jupyter notebook containing all code and markdown text;
2. `lastname_firstname_hw06.pdf`: a PDF compiled or exported from the executed notebook; and
3. `requirements.txt`: a short list of the packages needed to run the notebook, with version numbers.

Your notebook must run from top to bottom in a fresh session without manual edits. Use the supplied random seeds, and create each generator with `np.random.default_rng(seed)`. Keep written explanations concise, but explain the numerical behavior rather than reporting outputs alone.

Question 1: The nonparametric bootstrap and its limits (50 points)

Unless stated otherwise, draw a bootstrap sample by generating row indices with `rng.integers(0, n, size=(B, n))`, so that each row of the index matrix defines one bootstrap replicate.

1. **Standard error and bias.** Using seed 2026, generate $n = 25$ observations

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{LogNormal}(0, 1),$$

that is, $\log X_i \sim N(0, 1)$. These data are used again in Part 2.

- With seed 1 and $B = 5{,}000$, compute bootstrap replicates of both the sample mean and the sample median.
- For the **median**, report $\hat{\theta}$, $\widehat{\text{se}}_{\text{boot}}$, $\widehat{\text{bias}}_{\text{boot}}$, and the bias-corrected estimate $\hat{\theta}_{\text{bc}} = 2\hat{\theta} - \overline{\theta}^*$.
- For the **mean**, compare $\widehat{\text{se}}_{\text{boot}}$ with the usual formula $s/\sqrt{n}$. Show that the bootstrap standard error of the mean converges to $s/\sqrt{n}$ as $B \to \infty$, where $s^2 = n^{-1} \sum_i^n (X_i - \bar{X})^2$.

2. **Confidence intervals under skewness.** The target is the population mean $\theta = \mathbb{E}(X) = e^{1/2}$. Use $n = 25$ and seed 2027.

- For the data from Part 1, use the mean replicates you already have to compute the 95% normal, percentile, and basic bootstrap intervals.
- Estimate the coverage of the three procedures by simulation. Repeat $R = 1{,}000$ times: draw a new dataset of size n from LogNormal$(0, 1)$, then draw $B = 2{,}000$ bootstrap replicates of the mean **from the same generator**, and compute the three intervals. Report the empirical coverage, the Monte Carlo standard error of each coverage estimate, and the average interval length in one table.
- Explain the results. Which interval performs worst, and how is that related to the direction of skewness in the sampling distribution of $\bar{X}$?

Question 2: MCMC versus mean-field variational inference (50 points)

Effective sample size. Whenever an ESS is requested, use

$$N_{\text{eff}} = \frac{N}{1 + 2\sum_{k=1}^{K} \hat{\rho}_k},$$

where $\hat{\rho}_k$ is the lag-k sample autocorrelation of the retained draws and K is the last lag before the first negative $\hat{\rho}_k$. Report $\widehat{\text{MCSE}} = s_h/\sqrt{N_{\text{eff}}}$ alongside each MCMC posterior mean.

Model. Using seed 2468, generate $n = 20$ observations $Y_i \overset{\text{iid}}{\sim} N(2, 1.5^2)$. Then treat the data as observed and fit

$$Y_i \mid \mu, \tau \overset{\text{iid}}{\sim} N(\mu, \tau^{-1}), \qquad \mu \mid \tau \sim N\{m_0, (\kappa_0\tau)^{-1}\}, \qquad \tau \sim \text{Gamma}(a_0, b_0) \quad \text{(shape-rate)},$$

with $m_0 = 0$, $\kappa_0 = 1$, $a_0 = 2$, and $b_0 = 2$. This model is conjugate. With κ_n and m_n defined as in lecture,

$$a_n = a_0 + \frac{n}{2}, \qquad b_n = b_0 + \frac{1}{2}\sum_{i=1}^{n}(y_i - \bar{y})^2 + \frac{\kappa_0 n(\bar{y} - m_0)^2}{2\kappa_n},$$

the exact posterior has $\tau \mid y \sim \text{Gamma}(a_n, b_n)$ and

$$\mu \mid y \sim t_{2a_n}\left(m_n,\ \frac{b_n}{a_n\kappa_n}\right),$$

a Student t distribution with $2a_n$ degrees of freedom, location m_n, and squared scale $b_n/(a_n\kappa_n)$. Use these exact results only as a benchmark.

Gibbs sampling. Implement the two-block Gibbs sampler (i.e., iterate between μ and τ) using the full conditionals from lecture (slide 33). Start at $(\mu, \tau) = (0, 1)$, use seed 123, run 11,000 sweeps, and discard the first 1,000 as warm-up.

- Produce trace plots and autocorrelation plots, up to lag 40, for μ and τ.
- Report the posterior mean, posterior standard deviation, ESS, and MCSE for μ and τ, and a 95% credible interval for μ.
- Compare these results with the exact posterior.

Grading guide

Criterion	Points
Correct implementation and numerical results	50
Sound analysis and interpretation	25
Clear figures, tables, and concise communication	15
Reproducibility and complete submission	10
Total	**100**

Minor differences caused by software versions and Monte Carlo variability are acceptable. Numerical results without written interpretation will receive limited credit.